Computer Organization
And
Assembly Language

Lab Manual 09

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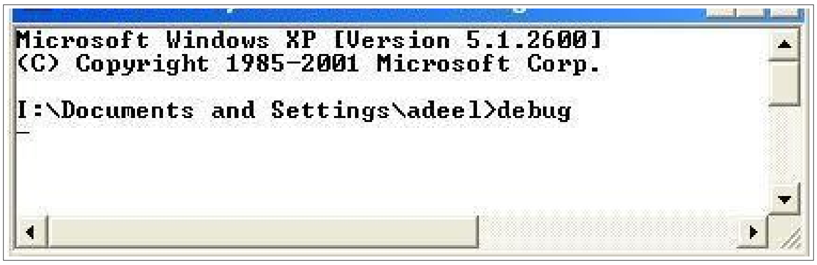
DEBUG provides **various** commands.

To invoke any of these commands, we enter the command character in either upper or lower case, and then enter any parameter that the command may require. Commands are executed when you press enter, not as you type them. You can change the line by pressing the backspace key to delete what you have typed and then type the correct characters. When the command line is as you want it, then press enter to execute the entire line.



Debug does not detect errors until you have pressed Enter. If you make a syntax error in an entry, the command line is displayed with the word error added at the point at which the error was detected. If a command line contains more than one error, only the first error is detected and indicated, and execution ceases at that point. Some of the commands are as following.

1. Display Help Screen (?)

2. Assemble (A)

3. Compare (C)

4. Dump (D)

5. Enter (E)

6. Fill (F)

7. GO (G)

8. Hexadecimal Arithmetic (H)

9. Input (I)

10. Load (L)

11. Move (M).

12. Name (N)

13. Output (O)14. Proceed (P), Trace (T)

15. Quit (Q)

16. Register (R)

17. Search (S)

18. Unassemble (U)

One of the most important commands is DUMP command.

Dump (D)

The Dump command is one we use often. It displays the contents of a series of memory locations. The syntax for this command is

-D *address1 address2*

We must specify address1, an optional starting address for the display, before you can specify address2, an optional ending address. If no addresses are specified, DEBUG starts displaying memory locations with DS:0100 or, (if Dump already has been used) with the byte following the last byte displayed by themost recent Dump command.

Dump always displays 16 bytes per line, beginning with the nearest paragraph (16-byte) boundary. This display rule may differ with the first and last lines displayed, because you may have asked DEBUG to start the dump with a memory location that was not on a paragraph boundary. If you do not specify an ending address, DEBUG always displays 128 bytes of memory. Each byte is shown in both Hexadecimal and ASCII representation.

```
I:\DOCUME~1\adeel>debug
-
-d
0B33:0100  B8 34 00 BB 43 00 01 D8-99 80 3E 20 99 00 75 24   .4..C.....> ..u$
0B33:0110  A2 24 99 0A C9 75 1D 0A-C0 74 19 8B 34 00 22 0B   .$...u...t..4.".
0B33:0120  13 B0 1A 06 33 FF 8E 06-00 96 F2 AE 07 75 05 4F   ....3........u.O
0B33:0130  89 3E 21 96 BB 06 97 80-3E 13 96 00 74 03 BB 4C   .>!.....>...t..L
0B33:0140  97 BE C3 96 8B 3E 05 98-B9 08 00 AC 3C 3F 75 02   .....>......<?u.
0B33:0150  8A 07 3C 20 74 01 AA 43-E2 F1 B1 03 B0 20 38 04   ..< t..C..... 8.
0B33:0160  74 12 B0 2E AA AC 3C 3F-75 02 8A 07 3C 20 74 01   t.....<?u...< t.
0B33:0170  AA 43 E2 F1 32 C0 AA C3-F6 46 04 02 75 43 8B D5   .C..2....F..uC..
-
-d 0010 001F
0B33:0010  97 05 17 03 97 05 5F 04-01 01 01 00 02 FF FF FF   ......_.........
-
-d 0010 L 10
0B33:0010  97 05 17 03 97 05 5F 04-01 01 01 00 02 FF FF FF   ......_.........
-
-
```

Lab Task

Unfortunately, some of your classmates' dump instruction is not working. You are supposed to help them by implementing such a program which works like dump instruction.

Implement a program which takes single address as input from user and displays 128 byte data from that address as dump instruction.

Note:

→ The user can enter address upto four characters, if less characters are entered, append remaining zeros. i.e. if user enters 2A, convert it to 002A.

→ Take base address from data segment address.